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SOLUTIONS GUIDE

FSK 116
Semester Test 2
Department of Physics, University of Pretoria

Date: 2024-05-14

Time: 3.0 hours

Total: 75 (max)

Examiner: RQ Odendaal

Moderator: Prof JM Nel

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Test register: Tear carefully on the line above and hand to the invigilator.

FSK 116: Semester Test 2

2024-05-14

Surname					Initials		
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Please read the following carefully.

1. Answer all the questions in the space provided on the question paper.
2. No pencil work will be marked. Do not use "tippex".
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues.
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams).
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations!**
A correct answer does not imply that you will receive full marks.
6. **Units must be used when numbers are substituted else marks will be subtracted.**
7. The use of the cheat sheet is only allowed subject to the rules stated on it.
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner.

Constants and formulas

$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$;
 $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; $N_A = 6.02 \times 10^{23} / \text{mol}$;
 $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 299\,792.5 \text{ km/s}$; $V_{\text{sphere}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$;
 $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = 5.97 \times 10^{24} \text{ kg}$; $\rho(\text{water}) = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = 6.37 \times 10^6 \text{ m}$; $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

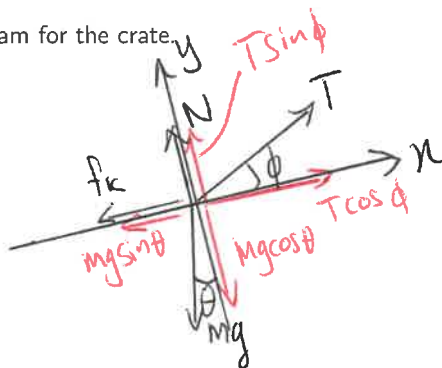
1 Question 1

[11]

An $m = 8.0 \text{ kg}$ crate is pulled a distance $d = 5.0 \text{ m}$ up a $\theta = 30^\circ$ incline by a rope angled at $\phi = 18^\circ$ above the incline. The tension in the rope is $T = 120 \text{ N}$, and the crate's coefficient of kinetic friction on the incline is $\mu_k = 0.25$.

(a) Sketch the free-body diagram for the crate.

(3)



(b) How much work is done by tension, by gravity and by the normal force?

(5)

$$W_T = T \cos \phi \times d = \dots = 0.57 \text{ kJ}$$

$$W_g = (-mg \sin \theta) d = \dots = -196 \text{ J}$$

$$W_N = 0$$

(c) What is the increase in the thermal energy of the crate and the incline?

(3)

$$\Delta E_{\text{thermal}} = \Delta E_{\text{int}} = f_k d$$

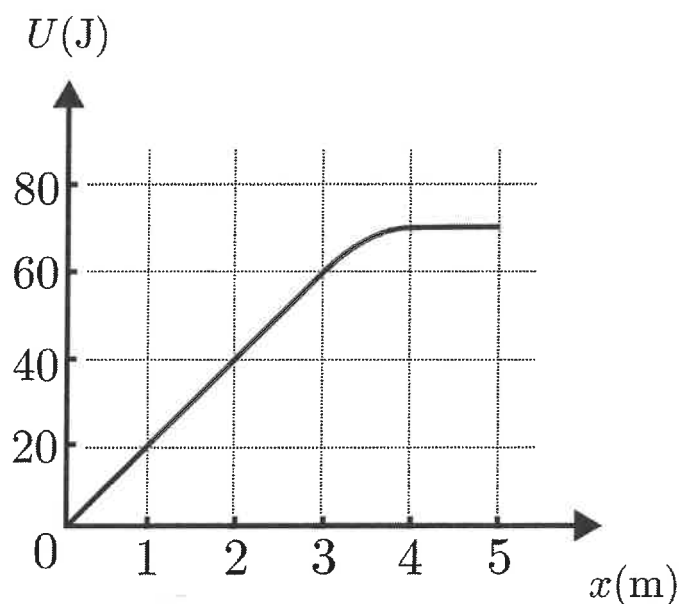
$$= \dots$$

$$= \mu_k (mg \cos \theta - T \sin \phi) d = \dots = 38.5 \text{ J}$$

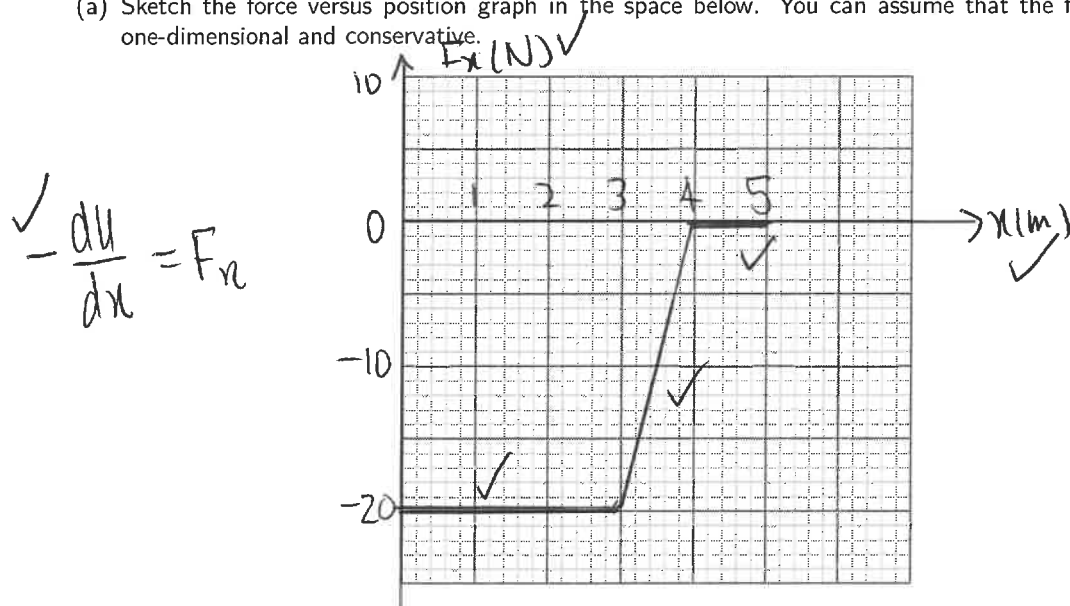
2 Question 2

[11]

A 100 g particle experiences a potential energy that changes as a function of the position x as depicted in the graph below.



(a) Sketch the force versus position graph in the space below. You can assume that the force, F_x , is one-dimensional and conservative. (6)



- (b) The particle is shot towards the right (the positive x -direction) from $x = 1.0$ m with a speed of 25 m/s. What is the particle's mechanical energy? (2)

$$K_{1.0m} = \dots = 31.25 \text{ J}$$

$$U_{1.0m} = 20 \text{ J (from graph)}$$

$$E_{\text{mech}} = U_{1.0m} + K_{1.0m} = \dots = 51.25 \text{ J}$$

- (c) Using the data provided, and your answer in (b), find where the particle has a turning point. (3)

$$U = 51.25 \text{ J} \checkmark \text{ since } K = 0 \checkmark \text{ at a turning point.}$$

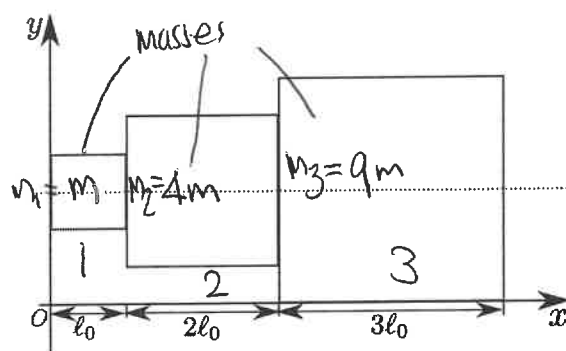
$$\text{From } U \text{ vs. } x \text{ graph this at around } x = (2.5 \text{ m}) \checkmark$$

$$x = 2.5 \text{ m exactly!}$$

3 Question 3

[6]

Three squares of side lengths ℓ_0 , $2\ell_0$, and $3\ell_0$ are placed next to each other with their centers along a straight line. What is the position of the center of mass of this configuration? Use the coordinate system as shown in the figure. You may assume that the squares are made of the same uniform material.



$$y_{\text{cm}} \text{ is on the line } y_{\text{cm}} = 1.5\ell_0$$

Find x_{cm} for each block, x_i

$$x_{\text{cm}} = \frac{1}{M} [m_1 x_1 + m_2 x_2 + m_3 x_3]$$

$$x_{\text{cm}} = 3.5\ell_0$$

$$\vec{r}_{\text{cm}} = (3.5\ell_0)\hat{i} + (1.5\ell_0)\hat{j}$$

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Question 4

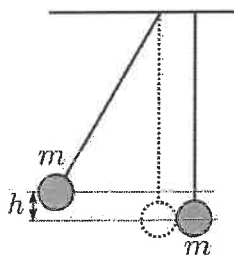
[10]

The pendulum on the left is pulled aside to the position shown in the figure (the dashed pendulum is the one on the left just before collision). It is then released and allowed to collide with another identical pendulum hanging vertically at rest. Assume the two balls at the ends of the pendulums have the same mass m and are made of a sticky material.

Before collision

$$U_{gi} = mgh \quad K_i = 0$$

$$U_{gf} = 0 \quad K_f = \frac{1}{2}mv^2$$



After collision

$$K_f = 0$$

$$U_{gf} = 2mgh'$$

$$K_i = \frac{1}{2}(2m)v'^2, \quad U_{gi} = 0$$

Unknown height

- (a) What is the speed of the ball on the left just before the collision. Show *all* assumptions made. (3)

Apply energy conservation

$$\Delta K + \Delta U = 0$$

$$K_f - K_i + U_{gf} - U_{gi} = 0$$

$$v = \sqrt{2gh}$$

- (b) After the collision, the two balls stick together. How high, in terms of h (the original height of the left ball), does the combination of two balls swing? (7)

Momentum Conservation

For collision: $\Delta \vec{p} = 0 \Rightarrow p_i = p_f$ (1-d)

$$mv = 2mv' \Rightarrow v' = \frac{v}{2}$$

Apply energy conservation:

$$\Delta K + \Delta U = 0$$

$$K_f - K_i + U_{gf} - U_{gi} = 0$$

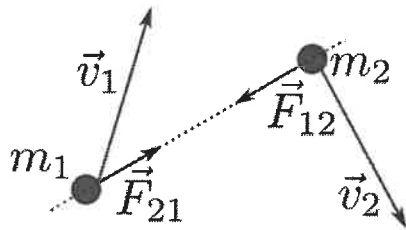
$$h' = \dots = \frac{h}{4}$$

5

Question 5

[6]

Two particles, particle 1 with mass m_1 and particle 2 with mass m_2 , in an **isolated** system exert forces $\vec{F}_{12}$ and $\vec{F}_{21}$ on one another. The two particles move with velocities $\vec{v}_1$ and $\vec{v}_2$ respectively. Prove that the total momentum of the system is a conserved quantity. Fully explain your notation as well as each step in your proof.

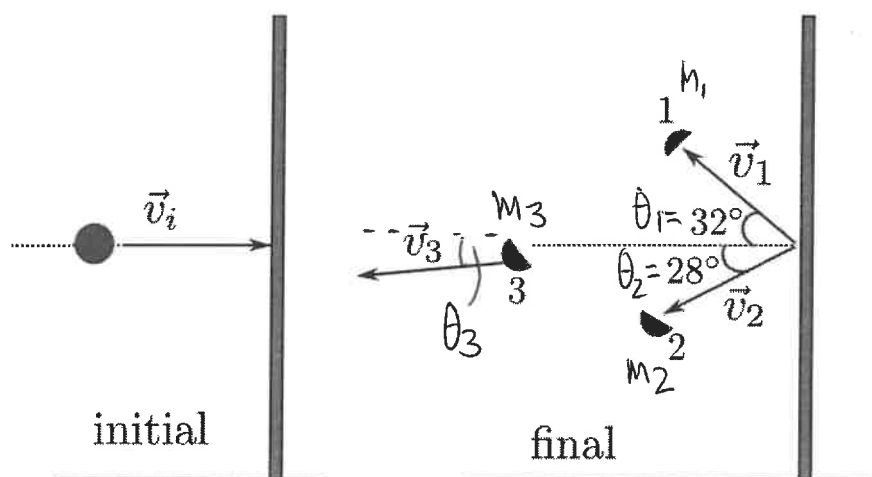


Proof is on page 211/212 of Serway
and in the class notes.

6 Question 6

[9]

A 6.00 kg clay ball is thrown directly against a perpendicular brick wall at a velocity of 22.0 m/s and shatters into three pieces, which all fly backward, as shown in the figure. The wall exerts a force on the ball of 2640 N for 0.100 s. One piece of mass 2.00 kg travels backward at a velocity of 10.0 m/s and at an angle of 32.0° above the horizontal. A second piece of mass 1.00 kg travels at a velocity of 8.00 m/s at an angle of 28.0° below the horizontal. What is the velocity of the third piece? Hint: The force only acts in the x -direction during the impact.



$$m_1 + m_2 + m_3 = M$$

$$m_3 = 3 \text{ kg}$$

$$\vec{I} = \vec{F} \Delta t = \Delta \vec{p}$$

In x -direction: $-F_x \Delta t = p_{xf} - p_{xi}$

$$-F_x \Delta t = [-m_1 v_1 \cos \theta_1 - m_2 v_2 \cos \theta_2 - m_3 v_{3x}] - M v_i$$

Solve for $v_{3x} = \dots = 35.992 \text{ m/s} = 36 \text{ m/s}$

In y -direction: $\Delta p_y = p_{yf} - p_{yi}$

$$0 = m_1 v_1 \sin \theta_1 - m_2 v_2 \sin \theta_2 + m_3 v_{3y} - 0$$

Solve for $v_{3y} = \dots = -2.28 \text{ m/s}$

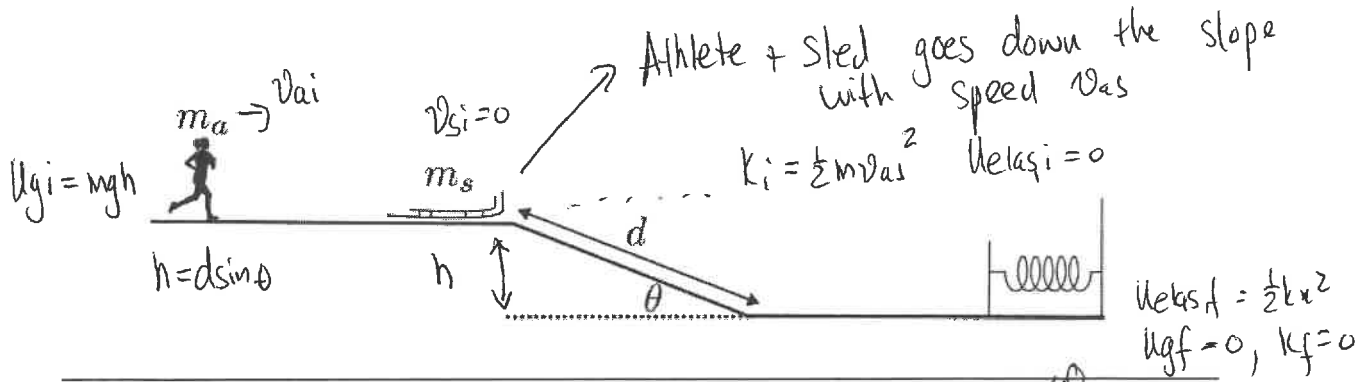
$$v_3 = \dots = 36.07 \text{ m/s}$$

$$\theta_3 = \dots = 3.6^\circ$$

[11]

Use the necessary conservation laws, and be clear what each physical quantity's value is initially and finally, even if the value is zero. You must also choose the necessary reference levels where applicable.

Question 7 continues on the next page.



Mom. conservation (perfect inelastic): $m_a v_{ai} + m_s v_{si} = (m_a + m_s) v_{as}$
 $v_{as} = \dots = 8 \text{ m/s}$

Energy conservation: $\Delta K + \Delta U_{grav} + \Delta U_{elas} = 0$

$$K_f - K_i + U_{gfs} - U_{gi} + U_{elsf} - U_{elas,i} = 0$$

$$0 - \frac{1}{2} m v_{as}^2 + 0 - mgd \sin \theta + \frac{1}{2} k x^2 - 0 = 0$$

Solve for $x = \dots = \sqrt{\frac{m v_{as}^2}{k} + \frac{2mgd \sin \theta}{k}}$

$$= \dots$$

$$= 3.46 \text{ m}$$

$$\approx 3.5 \text{ m}$$

8 Question 8

[5]

A force $\vec{F} = (12.0\hat{i} + 4.0\hat{j} - 16.0\hat{k}) \text{ N}$ is applied at the point A with coordinates $(-3.0 \text{ m}, 5.0 \text{ m}, 2.0 \text{ m})$. Find the torque of $\vec{F}$ about a rotation axis running through the point B with coordinates $(-3.0 \text{ m}, 2.0 \text{ m}, -6.0 \text{ m})$.

$$\vec{r} = \vec{r}_A - \vec{r}_B = \dots = (0\hat{i} + 3\hat{j} + 8\hat{k}) \text{ m}$$

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 \text{ m} & 3 \text{ m} & 8 \text{ m} \\ 12 \text{ N} & 4 \text{ N} & -16 \text{ N} \end{vmatrix} = \left\{ \hat{i}[(3)(-16) - (8)(4)] - \hat{j}[0(-96) + \hat{k}(0-36)] \right\} \text{ Nm}$$

$$= (-80\hat{i} + 96\hat{j} - 36\hat{k}) \text{ Nm}$$

9 Question 9

[7]

The flywheel of a steam engine starts to rotate from rest with a constant angular acceleration of $\alpha = 1.43 \text{ rad/s}^2$. The flywheel undergoes this constant angular acceleration for $t = 25.9 \text{ s}$ and then continues to rotate at a constant angular velocity, ω . After the flywheel has been rotating for 59.5 s, what is the total angle through which it has rotated since it started?

Flywheel starts at $\theta_i = 0$ with $\omega_i = 0$
accelerates for time t_a with constant angular acceleration

$$\Rightarrow \theta_a = \theta_i + \omega_i t_a + \frac{1}{2} \alpha t_a^2 = \dots = \frac{1}{2} \alpha t_a^2$$

Total time of rotation is t_b . For time $(t_b - t_a)$ it rotates through angle $\theta_b = \omega(t_b - t_a)$ but $\omega = \alpha t$, the final angular speed after time t_a

$$\Rightarrow \theta_b = \omega(t_b - t_a) = \alpha t_a(t_b - t_a)$$

$$\begin{aligned} \theta_{\text{total}} &= \theta_a + \theta_b \\ &= \frac{1}{2} \alpha t_a^2 + \alpha t_a(t_b - t_a) \\ &= \dots \\ &= 1724.07 \text{ rad} \\ &\approx 1724 \text{ rad. or } 1724. \end{aligned}$$

END OF TEST